Analyzing the SIR Model for Disease X

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Abstract

An infectious disease, called Disease X, is modeled using the SIR Model. The model has three different populations, each modeled by a differential equation: susceptible, infected, and recovered. The model incorporates parameters for infection rate, recovery rate, vaccination rate against and effectiveness, natural death and birth rates, and population size. Equilibrium and local analyses reveal two equilibrium points, whose behavior depends on parameter values. In this case, it was determined that there is one unstable saddle and one stable sink. Qualitative Analysis, including nullcline and vector field plots generated in MATLAB, confirmed these results. Numerical simulations further supported this conclusion.

1 Introduction

1.1 The SIR Model

The SIR Model is a system of differential equations used to model the spread of infectious diseases. At any given time, there are three populations: susceptible, infected, and recovered individuals. The SIR model has parameters that represent the rates of different events, such as transmission and exposure. These differential equations can be analyzed by equilibrium analyzing, quantitatively analyzing, and numerically solving the system.

1.2 Background

In this project, I will model a potentially deadly disease called Disease X. There is a vaccine for Disease X that is 97% effective given to 2% of the susceptible population at a given time. The infection rate upon exposure is 10%, while the recovery is 95%. The rate of death from infection is 1%. Once infected individuals recover, they are assumed to have permanent immunity. This model also accounts natural birth and death, both occurring at 0.2% at a time step, ensuring a constant total population.

1.3 The Model

Parameters

- $\alpha = 0.10$; the infection rate of susceptible individuals
- $\beta = 0.95$; the recovery rate of infected individuals
- $\delta = 0.01$; the rate of death given the disease

- $\epsilon = 0.97$; the rate of vaccination efficacy
- $\gamma = 0.02$; the rate of vaccination in the population
- $\mu = 0.002$; the rate of natural death and birth in the population
- $N = 10{,}000$; the number of people in the population

Equations

- $S(t)$ = susceptible individuals, who are neither infected nor recovered from the disease
- $I(t)$ = infected individuals
- $R(t)$ = recovered individuals
- $N(t) = S(t) + I(t) + R(t) = N$; the total population, which is constant

The Differential Equations
The following equations were formulated using the parameters and different events in the population.

$$\begin{cases} S'(t) &= \mu N - \alpha I(t)S(t) - \epsilon\gamma S(t) - \mu S(t) \\ I'(t) &= \alpha I(t)S(t) - \beta I(t) - \mu I(t) - \delta I(t) \\ R'(t) &= \beta I(t) + \epsilon\gamma S(t) - \mu R(t) \end{cases}$$

By Term:

- μN; the increase in susceptible individuals from natural birth.
- $\alpha I(t)S(t)$; the number of new infections. $I(t)S(t)$ represents the number of possible interactions between susceptible and infected populations. Reduces the susceptible population and increases the infected population.
- $\epsilon\gamma S(t)$; the number of people who are effectively vaccinated against the disease. Decreases the susceptible population and then increases the recovered population.
- $\mu S(t)$; the decrease in susceptible individuals who have a natural death.
- $\beta I(t)$; the number of infected individuals recovered at a given time. Reduces the infected population and increases the recovered population.
- $\mu I(t)$; the decrease in infected individuals who naturally die.
- $\delta I(t)$; the decrease in infected individuals who die from Disease X.
- $\mu R(t)$; the decrease in recovered individuals who naturally die.

2 Methods

2.1 Equilibrium Analysis

To perform an equilibrium analysis, all differential equations will be set equal to zero.

$$\begin{cases} S'(t) &= \mu N - \alpha I(t)S(t) - \epsilon\gamma S(t) - \mu S(t) = 0 \\ I'(t) &= \alpha I(t)S(t) - \beta I(t) - \mu I(t) - \delta I(t) = 0 \\ R'(t) &= \beta I(t) + \epsilon\gamma S(t) - \mu R(t) = 0 \end{cases}$$

The following equilibrium points are found.

$$P_1 = (\frac{\mu N}{\mu + \epsilon\gamma}, 0, \frac{\epsilon\gamma N}{\mu + \epsilon\gamma})$$

$$P_2 = (\frac{\beta + \mu + \delta}{\alpha}, \frac{\mu N}{\beta + \mu + \delta} - \frac{\epsilon\gamma + \mu}{\alpha}, \frac{\beta N}{\beta + \mu + \delta} + \frac{\beta(\epsilon\gamma + \mu) - \epsilon\gamma(\beta + \mu + \delta)}{\alpha\mu})$$

This equilibrium point P_1 represents a disease-free state, where no infections exist in the population. The equilibrium point P_2 represents and endemic equilibrium in which infection exists at a constant level.

Local Analysis

Using local analysis, the behavior of the equilibrium points will be determined. To simplify the analysis, the system is simplified to two dimensions by rearranging the total population equation to be:

$$R = N - S - I$$

This allows the behavior of the system to be studied using only the equations $S'(t)$ and $I'(t)$, without loss of generality.

$$\begin{cases} S'(t) &= \mu N - \alpha I(t)S(t) - \epsilon\gamma S(t) - \mu S(t) \\ I'(t) &= \alpha I(t)S(t) - \beta I(t) - \mu I(t) - \delta I(t) \end{cases}$$

The Jacobian matrix of the system will be computed:

$$J = \begin{bmatrix} \frac{\partial S'}{\partial S} & \frac{\partial S'}{\partial I} \\ \frac{\partial I'}{\partial S} & \frac{\partial I'}{\partial I} \end{bmatrix}$$

Substituting the equations,

$$J = \begin{bmatrix} -\alpha I - \epsilon\gamma - \mu & -\alpha S \\ \alpha I & \alpha S - \beta - \mu - \delta \end{bmatrix}$$

Equilibrium Point P_1 :

At the equilibrium point $(\frac{\mu N}{\mu+\epsilon\gamma}, 0, \frac{\epsilon\gamma N}{\mu+\epsilon\gamma})$, the Jacobian matrix becomes:

$$J_1 = \begin{bmatrix} -\epsilon\gamma - \mu & -\alpha\frac{\mu N}{\mu+\epsilon\gamma} \\ 0 & \alpha\frac{\mu N}{\mu+\epsilon\gamma} - \beta - \mu - \delta \end{bmatrix}$$

To determine the behavior, the eigenvalues must now be derived using the following equation.

$$det(J_1 - \lambda I) = 0$$

The eigenvalues are:

$$\lambda_1 = -(\epsilon\gamma + \mu)$$

$$\lambda_2 = -(\beta + \mu + \delta) + \frac{\alpha\mu N}{\mu + \epsilon\gamma}$$

λ_1 will always be negative. The sign of λ_2 depends on the value of the parameters and will change the behavior around this equilibrium point depending on the sign.

To account for this, the basic reproduction number must be found. The Basic Reproduction Number (R_0) is used to measure the contagiousness of the disease [2]. It is used in this case because the behavior of the equilibrium is subject to change depending on the value of the parameters. When $R_0 = 1$, this signals that the behavior of the system is changing [1]. To find R_0, the equation for λ_2 is set equal to 0 and the equation is manipulated to equal 1.

$$-(\beta + \mu + \delta) + \frac{\alpha\mu N}{\mu + \epsilon\gamma} = 0$$

$$(\beta + \mu + \delta) = \frac{\alpha\mu N}{\mu + \epsilon\gamma}$$

$$1 = \frac{\alpha\mu N}{(\beta + \mu + \delta)(\mu + \epsilon\gamma)}$$

Therefore,

$$R_0 = \frac{\alpha\mu N}{(\beta + \mu + \delta)(\mu + \epsilon\gamma)}$$

When $R_0 > 1$, $\lambda_2 > 0$, so the equilibrium is an unstable saddle . When $R_0 < 1$, $\lambda_2 < 0$, so the equilibrium is a stable sink.

Equilibrium Point P_2:

$$J_2 = \begin{bmatrix} -\alpha I - \epsilon\gamma - \mu & -\alpha S \\ \alpha I & 0 \end{bmatrix}$$

Since the expressions for S and I are lengthy, it is more efficient to determine the system's behavior by computing the trace and determinant of the Jacobian.

$$Trace(J_2) = -\alpha I - \epsilon\gamma - \mu = \frac{-\mu N}{\alpha(\beta + \mu + \delta)}$$

$$Det(J_2) = \alpha^2 SI = \mu N - (\epsilon\gamma + \mu)\frac{\beta + \mu + \delta}{\alpha}$$

The Jacobian trace will always be negative since the parameters are always positive, and the Jacobian determinant is dependent on the values of the parameters. When $R_0 > 1$, the determinant is positive, which means the equilibrium is a stable sink. When $R_0 < 1$, the determinant is negative, which means the equilibrium is an unstable saddle.

2.2 Qualitative Analysis

To study the system qualitatively, nullclines are computed by setting $S'(t) = 0$ and $I'(t) = 0$. These equations are written in terms of S and I and their intersections identify equilibrium points.

$$\begin{cases} S'(t) &= \mu N - \alpha I(t)S(t) - \epsilon\gamma S(t) - \mu S(t) = 0 \\ I'(t) &= \alpha I(t)S(t) - \beta I(t) - \mu I(t) - \delta I(t) = 0 \end{cases}$$

The nullclines are:

$$S = \frac{\beta + \mu + \delta}{\alpha}$$
$$S = \frac{\mu N}{\alpha I + \epsilon\gamma + \mu}$$
$$I = 0$$

Substituting the parameters, the equations become

$$S = 9.62$$
$$S = \frac{20}{0.1I + 0.0214}$$
$$I = 0$$

These nullclines intersect at $(9.62, 0)$ and $(9.62, 20.57)$, which matches the equilibrium points found in the previous section. The nullclines are graphed using MATLAB.

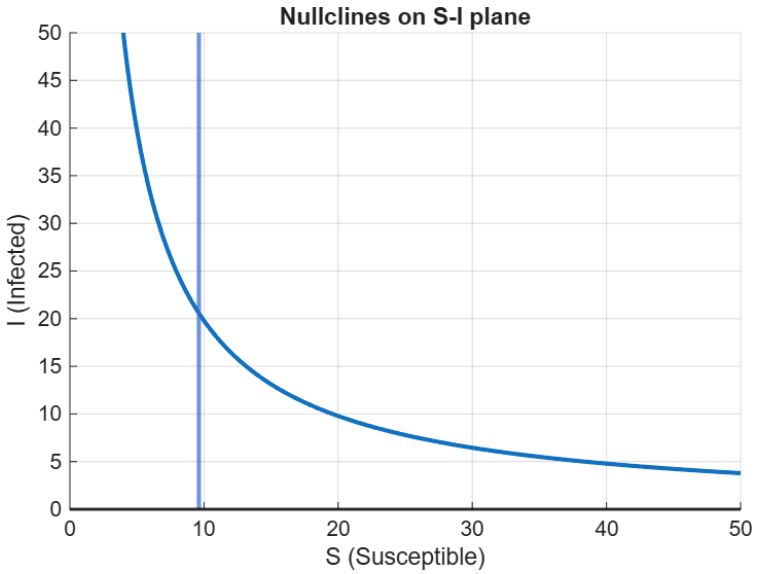


Figure 1: S-nullclines and I-nullcline

To show the behavior of the system, the vector field is put over the graph above.

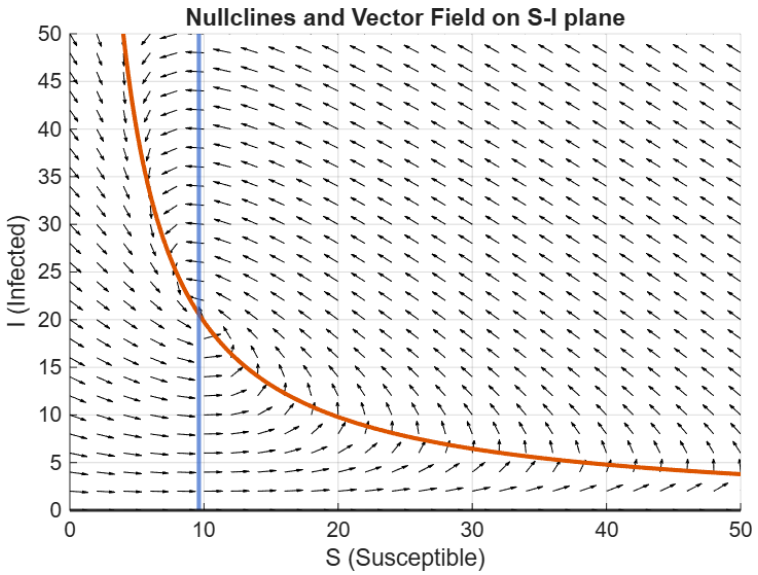


Figure 2: Nullclines and Vector Field

Looking at this graph, the point at $(9.62, 20.57)$ is clearly a stable sink and the point $(9.62, 0)$ there is an unstable saddle. Since the system is behaving this way, this means $R_0 > 1$. To verify this conclusion, the value of R_0 is found to be $R_0 \approx 97.1$. Since $R_0 > 1$, Disease X will persist in this population.

2.3 Numerical Analysis

The conclusions in the previous sections can be verified by solving the system numerically. The solutions will be solved and graphed using MATLAB. First, a simulation will be made with the parameters given in the problem. For each equilibrium point, two points will be chose near the equilibrium to be initial points of the simulated system. The numerical solutions will be overlaid on Figure 2 to demonstrate the behavior.

Testing points near P_1

At (9, 0.1, 9990.99):

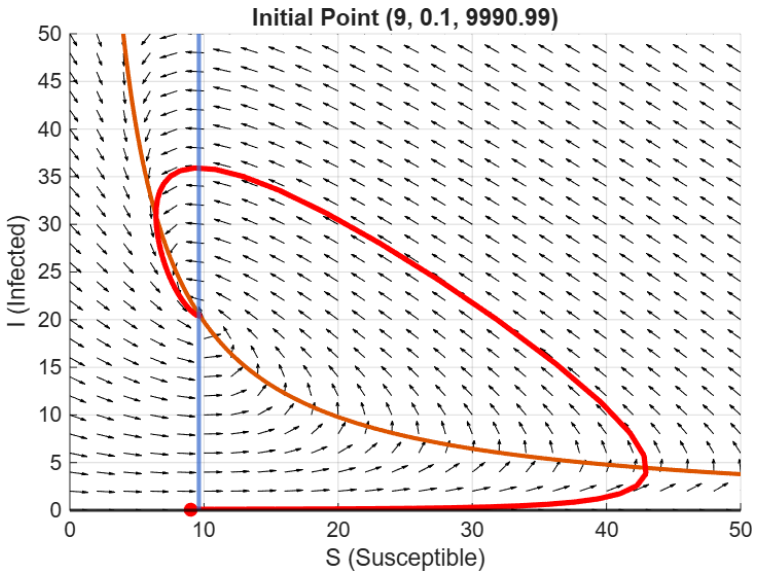


Figure 3: Numerical Solution with Initial Point (90, 0.1, 9990.99)

Although the infected population is extremely small, the population does not disappear. It shifts away from P_1 shown on the graph, which supports the fact that P_1 is an unstable saddle. It shifts towards P_2 and reaches this equilibrium.

At (10, 0.5, 9989.5):

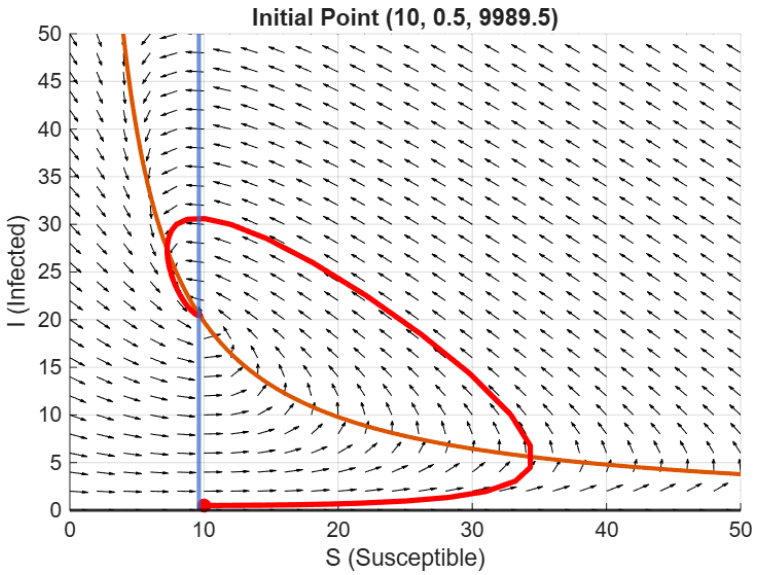


Figure 4: Numerical Solution with Initial Point (10, 0.5, 9989.5)

This is very similar to the first test point. Having a small initial infection population does not mean that the disease will die out. It moves away from the P_1 equilibrium towards the P_2 equilibrium.

Testing points near P_2

At (20, 25, 9955):

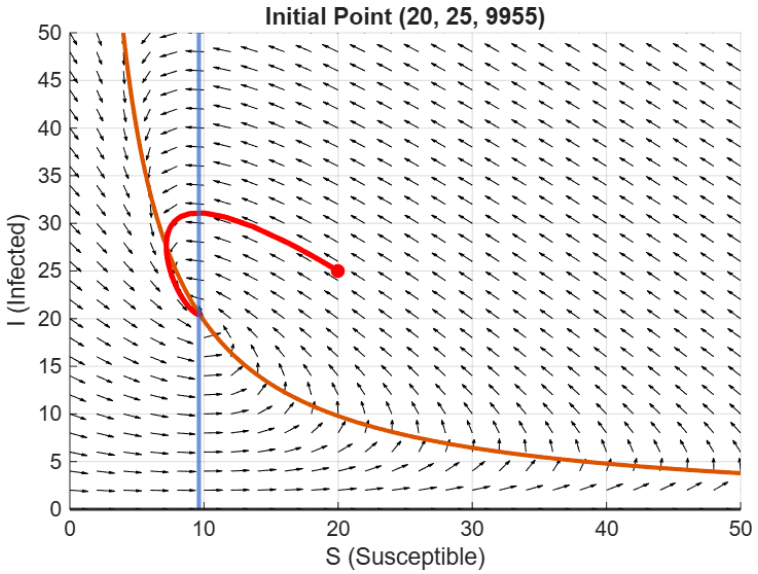


Figure 5: Numerical Solution with Initial Point (20, 25, 9955)

These graphs show the populations stabilizing towards a steady state. It confirms that P_2 is a stable sink. The disease continues in the population at a constant level rather than disappearing or growing indefinitely.

At (5, 45, 9950):

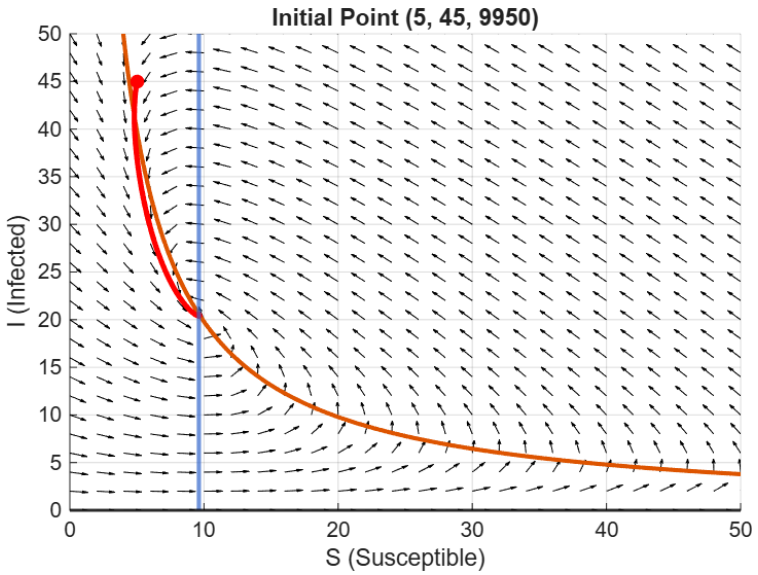


Figure 6: Numerical Solution with Initial Point (5, 45, 9950)

Even at a higher initial infected population, the populations stabilize at the same equilibrium

as the previous test point. It reinforces P_2 is the stable equilibrium when $R_0 > 1$. Disease X will not die out in this population.

3 Results

The equilibrium analysis found two equilibrium points: $P_1 = (\frac{\mu N}{\mu+\epsilon\gamma}, 0, \frac{\epsilon\gamma N}{\mu+\epsilon\gamma})$ and $P_2 = (\frac{\beta+\mu+\delta}{\alpha}, \frac{\mu N}{\beta+\mu+\delta} - \frac{\epsilon\gamma+\mu}{\alpha}, \frac{\beta N}{\beta+\mu+\delta} + \frac{\beta(\epsilon\gamma+\mu)-\epsilon\gamma(\beta+\mu+\delta)}{\alpha\mu})$. For the given parameter values, P_1 is an unstable saddle and P_2 is a stable sink. Qualitative analysis using nullclines and vector fields supports this conclusion about the behavior. Numerical simulations further confirm that trajectories move away from P_1 and converge to P_2 demonstrating the persistence of Disease X in this population.

4 Conclusion

This project analyzed the spread of Disease X using an SIR model with vaccination and demographic effects. Two equilibrium points were identified, and their stability depends on the basic reproduction number R_0. Since $R_0 > 1$ for the given parameters, the disease-free equilibrium is unstable while the endemic equilibrium is stable. Both qualitative and numerical analyses support this conclusion, showing that the disease will persist in the population. Future work could explore how changes in vaccination rates, population size, or additional factors would influence disease spread dynamics.

Statement

Mathematica was used to solve the system of equations. MATLAB was used to produce all the plots. ChatGPT was used to help generate and edit the code for the MATLAB plots, and it was also used to help with some formatting issues with LaTeX.

Acknowledgement

I would like to acknowledge my peer who anonymously reviewed my work. They gave useful advice that helped me clarify my local analysis.

References

[1] F. S. Alshammari and M. A. Khan. Dynamic behaviors of a modified sir model with nonlinear incidence and recovery rates. *Alexandria Engineering Journal*, 60(3):2997–3005, 2021.

[2] Klaus Dietz. The estimation of the basic reproduction number for infectious diseases. *Statistical Methods in Medical Research*, 2(1):23–41, 1993.